

Joel Hempel

Machine Learning and Full-Stack Developer

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Education

BSc, Computer Science
University of British Columbia
• **GPA:** 4.2/4.33

Vancouver, Canada
Nov 2025

Experience

Software Developer
Checker Software

Vancouver, BC
05/2025 – Present

- **Shipped 20+ full-stack features/bug fixes and reviewed 20+ pull requests for Diffchecker.com** (≈ 1 million monthly users) in the first 7 months, boosting UI/UX quality and catching critical ML-related regressions.
- **Developed in-house image-alignment algorithm** using computer vision (homography).
- **Completely overhauled** AI-powered explain feature creating a custom algorithm with AI-agents fully surpassing previous solutions.
- **Technologies:** Next.js, React, TypeScript, PostgreSQL, Docker, GitHub Actions, Gemini, OpenCV

Software Team Lead (Computer Vision & WebDev)
UBC Subbots

Vancouver, BC
09/2024 – Present

- **Reached semi-finals** during 2025 RoboSub competition.
- Recruited and led a **12-member cross-functional team**, overseeing project planning, task delegation, and peer training to accelerate development and improve collaboration.
- **Designed firmware and software for simulation, controls, and inertial measurement unit (IMU).**
- Implemented **trajectory-generation algorithms** to guide the AUV around buoys.
- **Created and deployed new website within 1-month**
- **Technologies:** C++, ROS2, Gazebo, Arduino, Ubuntu

SWE Intern (Machine Learning & Full-Stack)
Metrizd Consulting Inc.

Vancouver, BC
09/2023 – 08/2024

- **Built and deployed an alpha version of a computer vision powered cost-estimation web-application** to reduce contractor estimation time by around 80%.
- Co-created a distributed back-end solution for separate training, inference, and core servers: allowing for easier scaling.
- Engineered a ground-truth dataset of **10,000 images**, reduced synthetic data generation/augmentation time by **95%**, and trained a **top-performing computer vision model with 95% test-set accuracy**.
- Authored comprehensive documentation (on **20,000+** lines of code across 4 repositories), streamlining new-hire onboarding.
- Expanded an existing **front-end by adding new 10 new features** and improving user-experience.
- Proposed a **breakthrough feature** for AI-assisted symbol labeling **increasing accuracy by 400%**.
- Conducted experimental research on language models for potential future products.
- **Technologies:** Python, PyTorch, Next.js, AWS, Supabase, JavaScript, YOLO, SAM, LLMs

Undergraduate Research Assistant
Systopia Lab

Vancouver, BC
05/2023 – 12/2023

- Researched spring-onset detection using satellite data and flower proxies combined with machine learning.
- Developed a data pre-processing pipeline to transform temperature data and built the highest-performing model for the paper **reducing the in-region error of the state-of-the-art first-bloom index by 16.9%**.
- **Co-authored and presented a paper at NeurIPS 2023**
- **Technologies:** Python, scikit-learn

Technologies

Languages: Python, Java, C, C++, JavaScript, TypeScript, HTML, CSS, Erlang
Frameworks & Libraries: PyTorch, React, Next.js, FastAPI, CUDA, OpenAI, Gemini
Web/Databases: Supabase, SQL, Tailwind, PHP, PostgreSQL, Node.js
Tools: Git, Docker, AWS, Google Cloud, JUnit, Google Colab
ML: Traditional Models, Neural Networks, Computer Vision, LLMs